Does a paraglider move differently in a group?

Group time against solo flight, on open flights

3 November 2026

Four questions

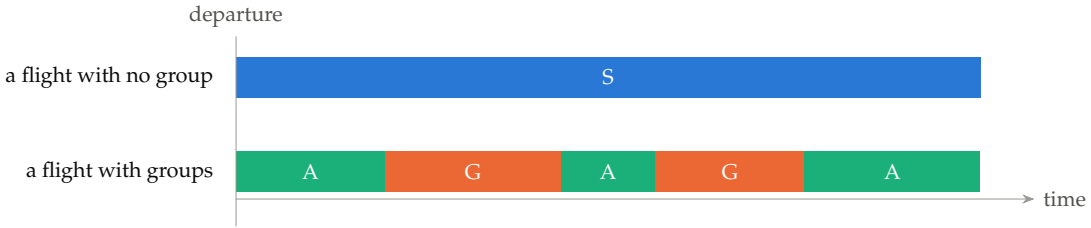
1 Do flights in a group move differently from solo flights?
Raw

2 With route, season, hour and wing fixed, does the group still make a difference?
All fixed

3 How much does each of these factors count?
Each factor

4 Does the same flight change when it enters a group, and when it leaves?
Same flight

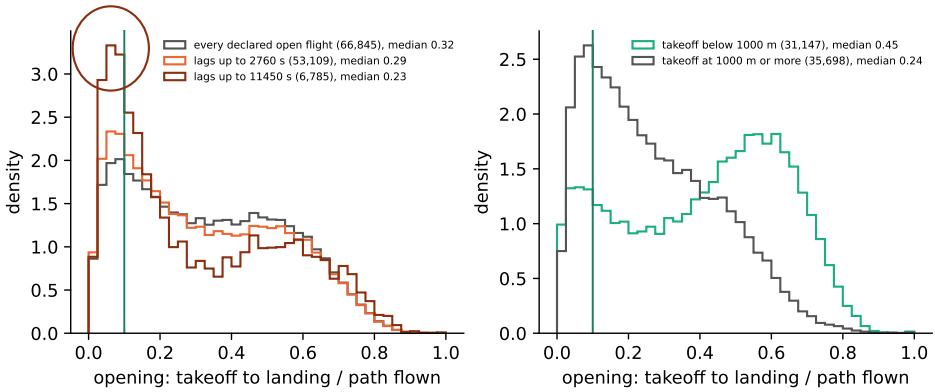
Three populations of flight time



- S (Solo flight): flights that never entered a group.
- G (Group flight): group time of the flights that entered one.
- A (Solo stretch): the solo part of the group flights.

A *piece* is an uninterrupted stretch of one population.

Which flights: the open ones

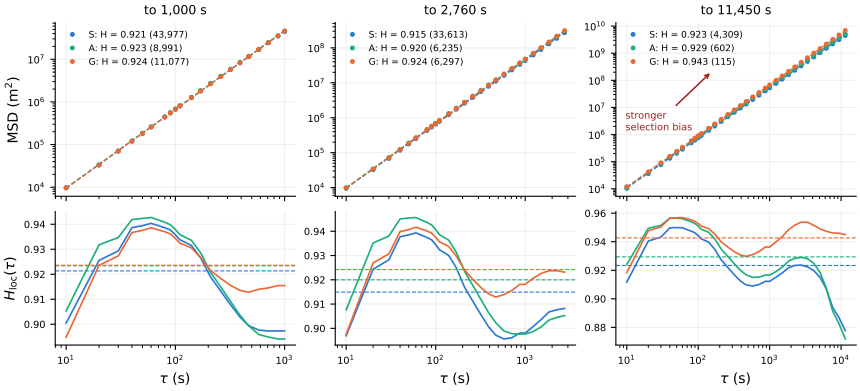


- **Opening** $= \dfrac{\|\mathbf{r}_{\text{land}} - \mathbf{r}_{\text{take}}\|}{\sum_i \|\mathbf{r}_{i+1} - \mathbf{r}_i\|}$: 1 straight, ≈ 0 back at the takeoff.
- We keep opening ≥ 0.1: 84% of the flights; 88% of low takeoffs, 80% of high ones (right).

Question 1

Do flights in a group move differently from solo flights?

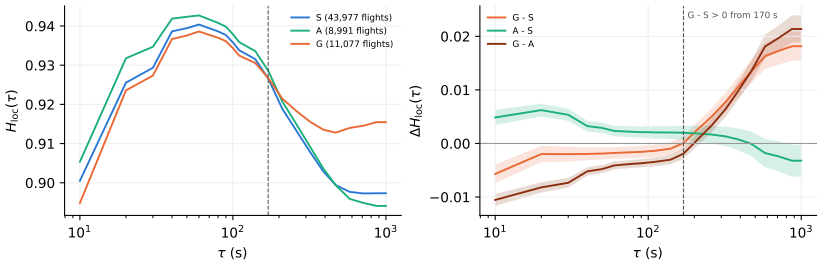
Fitted H and H_{loc} of S, A and G



- MSD $\sim \tau^{2H_{\mathrm{loc}}}$.
- Each fit on the pieces of at least end lag / 0.8: **flights constant along the lags, fewer for longer fits**.

	to 1000 s	to 2760 s	to 11 450 s
$H_{\mathrm{G}} - H_{\mathrm{S}}$	$+0.002$ $[+0.001, +0.004]$	$+0.009$ $[+0.007, +0.011]$	$+0.019$ $[+0.006, +0.032]$

Raw: group time is straighter at the long lags



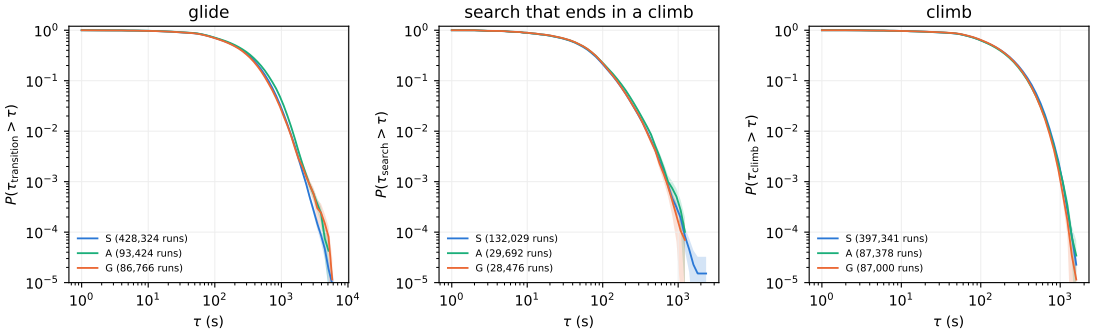
- $H_{\mathrm{loc}}(\tau)$: local exponent of the MSD $\sim \tau^{2H_{\mathrm{loc}}}$.
- $H_{\mathrm{loc}}(1000\ \mathrm{s})$: G 0.915, S 0.897, A 0.894.
- $\mathrm{G} - \mathrm{S} = +0.018\ [+0.015, +0.021]$; positive from 170 s

Same flights at every lag: pieces of at least 1250 s, so lags up to 1000 s.

At first the group waits to form, a solo pilot leaves at once. After one glide (median $\approx$ 200 s) group time gains, probably because group flights favour zones and days with more thermals.

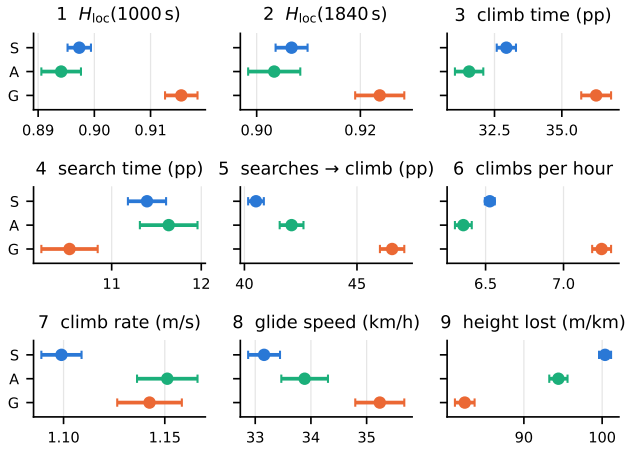
→ how we compute H_{loc} → the bands → longer lags, other pools → map of flights

Raw: the durations of the phases



- Climbs, searches and glides last as long in S, A and G: the survival curves overlap.

Raw: group time also flies differently



1, 2 G − S ≈ 0.02, with pieces of at least 1250 s (1) or 3450 s (2).

3, 6 G climbs more and finds more climbs.

4, 5 G searches less and better.

7 A ≈ G: zones with stronger thermals.

8 G keeps a higher glide speed (Williams et al. 2018).

9 G glides flatter.

pp: percentage points; 90% bands.
G − S excludes 0 in panels 3–9.

Flights S / A / G: panel 1, 43 977 / 8 991 / 11 077; panel 2, 33 613 / 6 235 / 6 297; panels 3–9 (with phase time), 27 665 / 6 397 / 7 896.

→ how they are computed
→ how each flight counts

Question 2

With route, season, hour and wing fixed,
does the group still make a difference?

The raw gap mixes everything

Group time differs from solo time in

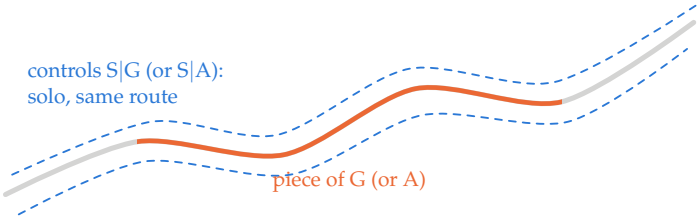
- the **route** flown,
- the **atmospheric conditions** (the season and the hour),
- the **wing**,
- the pilots who fly in groups.

We take them apart

1. fix them, and see what is left;
2. split the gap among them.

Same route, season, hour and wing: does the group still make a difference?

The controls S|G and S|A: same route, season, hour and wing



route: 80% of the piece within 500 m

another date, ±2 months

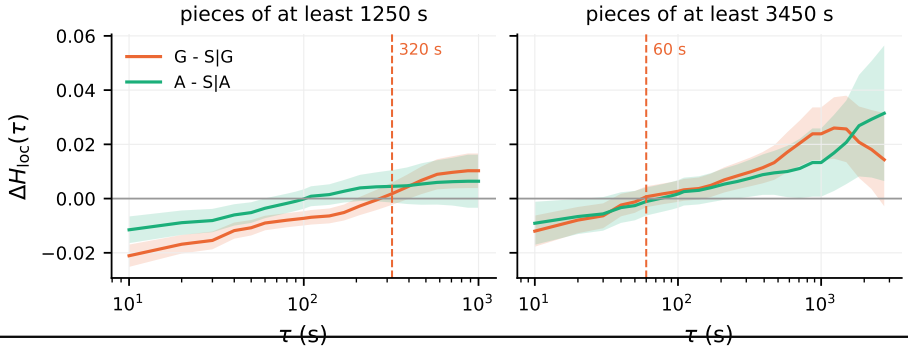
hour of the day ±2 h

same wing class

- S|G answers: **how does a solo pilot fly the stretch that** G **flew?** For each piece of G, the segments of solo flights along the same route, same direction. S|A: the same for a piece of A.
- A segment is kept only if it meets the four conditions on the right.
- Tolerances chosen to keep at least 200 pieces of G per wing level (beg/exp); A keeps 123 beginners on long pieces.

→ what is a control → why the long dataset stops at 2760 s

On the same route, group pilots are straighter, in a group or alone

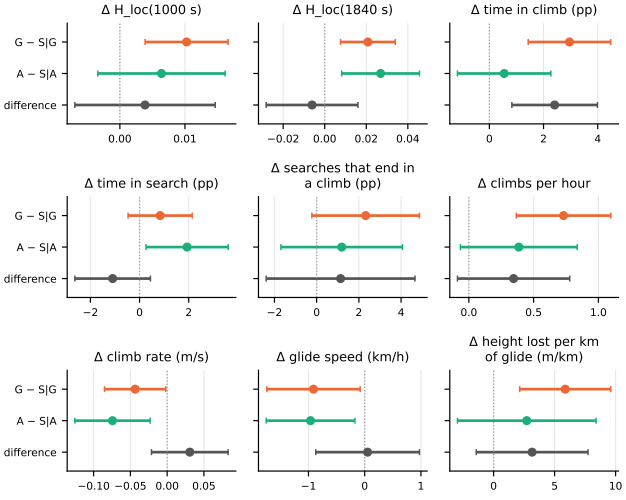


90% bands	$H_{\mathrm{loc}}(1000\ \mathrm{s})$	$H_{\mathrm{loc}}(1840\ \mathrm{s})$
G − S|G = the group + pilots and day	+0.010 [+0.004, +0.017]	+0.021 [+0.008, +0.034]
A − S|A = the same pilots and days, without the group	+0.006 [−0.003, +0.016]	+0.027 [+0.008, +0.046]
difference = the group alone, ≈ 0	**+0.004 [−0.007, +0.015]**	**−0.006 [−0.028, +0.016]**

Pieces of G with a control: 2 717 (from 2 474 flights, 1 378 pilots) at ≥ 1250 s, 700 at ≥ 3450 s; of A: 1 607 (1 504 flights, 990 pilots) and 338. Controls: 8 094 for G, 4 522 for A.

→ why the difference → H_{loc} curves → fitted H → robustness

Same route: net of pilots and day, only the time in climb remains



Pieces of G / A: 2 717 / 1 607 for $H_{\mathrm{loc}}(1000\,\mathrm{s})$, 700 / 338 for $H_{\mathrm{loc}}(1840\,\mathrm{s})$, 1 422 / 831 for the phase statistics.

- Panels 1–2: at 1840 s $\mathrm{A}-\mathrm{S}|\mathrm{A} \approx \mathrm{G}-\mathrm{S}|\mathrm{G}$. The gap comes with the pilots and the day, not with the group; to fix them, the same flight (question 4).
- Only the time in climb survives in the difference: $+2.4$ pp $[+0.8, +4.0]$.
- Glide speed: raw $+2.1$ km/h, here -0.9 $(\mathrm{G}-\mathrm{S}|\mathrm{G})$ and -1.0 $(\mathrm{A}-\mathrm{S}|\mathrm{A})$. **Probably group flights are on routes where everyone glides faster.**

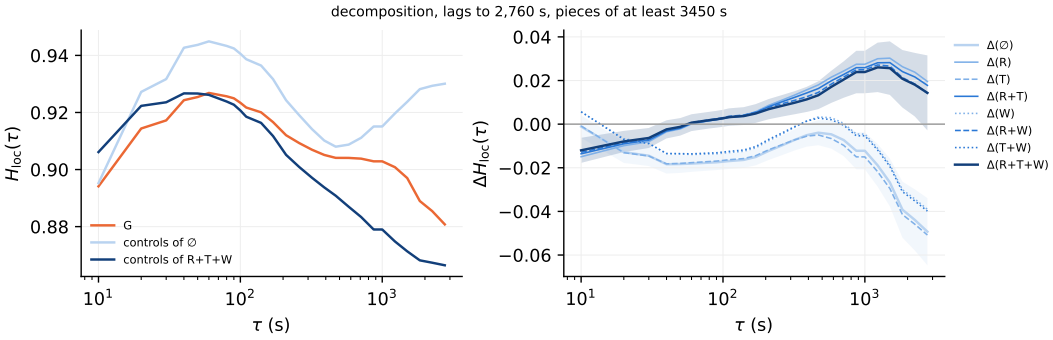
90% bands.

→ by wing level → robustness

Question 3

How much does each of these factors count?

The eight sets of controls: the route splits them in two

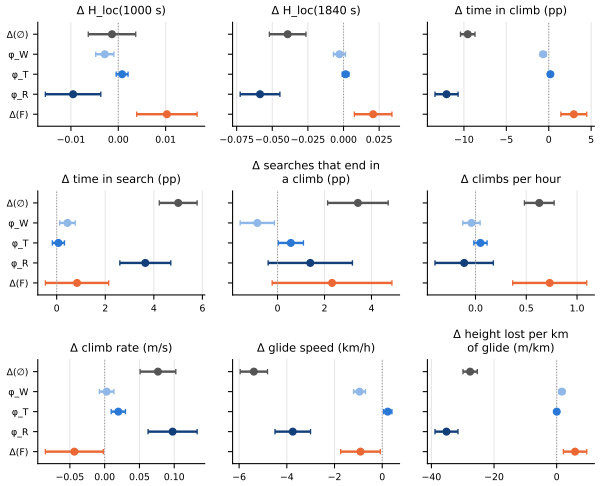


- Right: $\Delta H_{\mathrm{loc}}(\tau)$ of G minus eight sets of controls, one per subset of route R, season and hour T, wing W fixed, from ∅ to $R{+}T{+}W = \mathrm{S}|\mathrm{G}$.
- With R (solid): positive at long lags. Without R: negative. T, W: little.

∅ is not the raw gap: raw $+0.017$, here -0.039 at 1840 s, because the controls are windows of very long solo flights, which are straighter. Bands for ∅ and $R{+}T{+}W$. Left: H_{loc} of G, of the controls of ∅ and of S|G. 700 pieces of G of at least 3450 s, the same in every set.

→ the nine quantities, three sets

How much each factor explains: the Shapley split



Pieces of G, the same in every set: 2 717 for $H_{\mathrm{loc}}(1000\ \mathrm{s})$, 700 for $H_{\mathrm{loc}}(1840\ \mathrm{s})$, 1 422 for the phase statistics.

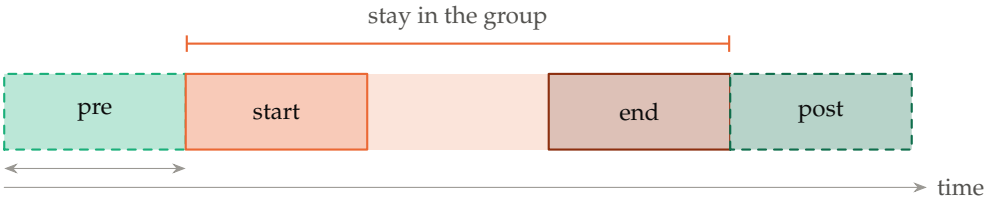
- **Question:** how much of the move from $\Delta(\varnothing)$ to $\Delta(F) = \mathrm{G} - \mathrm{S}|\mathrm{G}$ comes from each of R, T, W?
- **Method:** the gap is computed for each combination of fixed factors; we average the change from adding k over the 6 orders. φ_k = **minus that average.**
- **They add up:** $\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W$. Time in climb: $-9.5 = +3.0 - 12.0 + 0.2 - 0.7$.
- **The route carries most of the move:** 98% at $H_{\mathrm{loc}}(1840\ \mathrm{s})$, 96% time in climb, 84% glide speed.

→ how φ is computed → why they add up

Question 4

Does the same flight change when it enters a group, and when it leaves?

The same flight, before, during and after



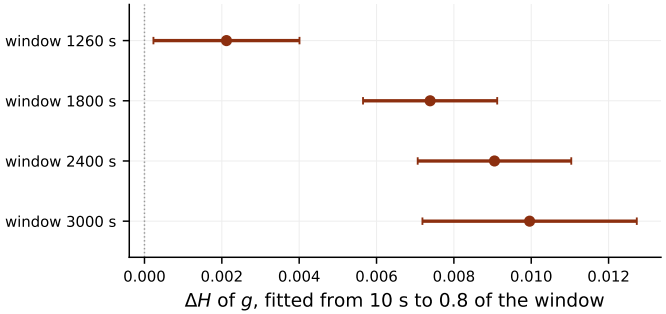
$$D_{\text{in}} = \text{start} - \text{pre}, \quad D_{\text{out}} = \text{end} - \text{post}, \quad g = \tfrac{1}{2}(D_{\text{in}} + D_{\text{out}}), \quad d\,w = \tfrac{1}{2}(D_{\text{in}} - D_{\text{out}})$$

- Same pilot, glider, logger and day: it removes who flies and when*.
- g: what the group does; $d\,w$: the drift of the flight over time.

*Only in part: the windows can be hours apart and over different terrain, so the weather may change.

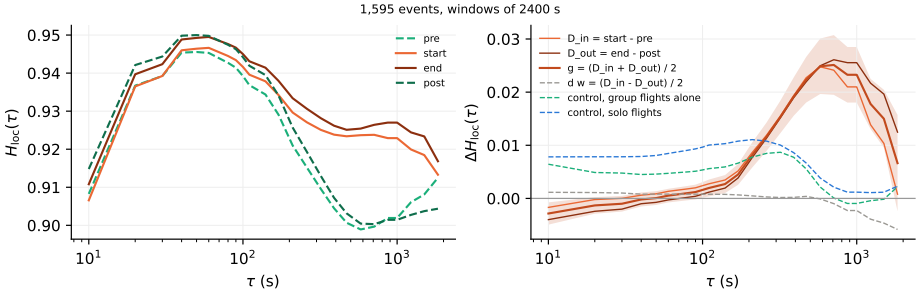
Fitted H in the same flight

ΔH **of** g. $\frac{1}{2}\left[(H_{\text{start}} - H_{\text{pre}}) + (H_{\text{end}} - H_{\text{post}})\right]$



ΔH of g is positive for every window (fit from 10 s to 0.8 of the window).

With windows of 2400 s the flight is straighter in the group

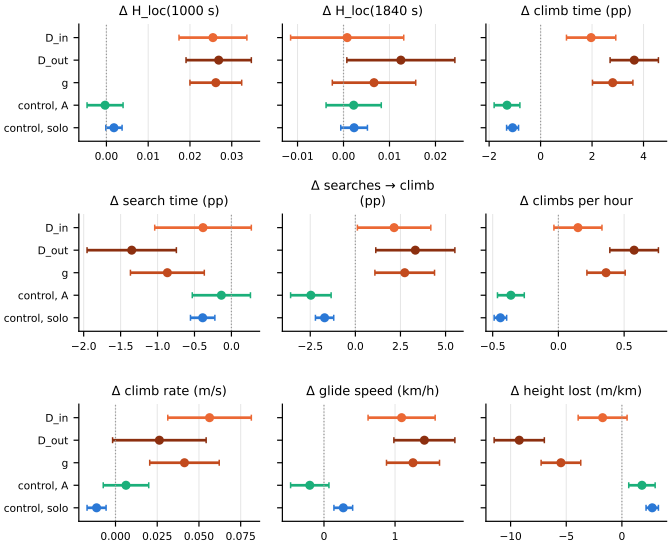


- $g = +0.026\ [+0.020, +0.032]$ at $H_{\mathrm{loc}}(1000\ \mathrm{s})$.
- Time in climb $+2.8\ [+2.0, +3.6]$ pp.
- $g > 0$ from 50 s, as $\mathrm{G} - \mathrm{S} \mid \mathrm{G}$ on long pieces (60 s); raw $\mathrm{G} - \mathrm{S}$: 170 s.
- The controls: how much does H_{loc} change between two consecutive windows, with no change of state?

1 595 events from 1 542 flights. Control curves: about 5 300 pairs in the solo time of group flights, 26 600 in solo flights, one random pair of consecutive windows per solo stretch of at least 4800 s; not matched to the events.

→ all the differences

The same flight: all the quantities

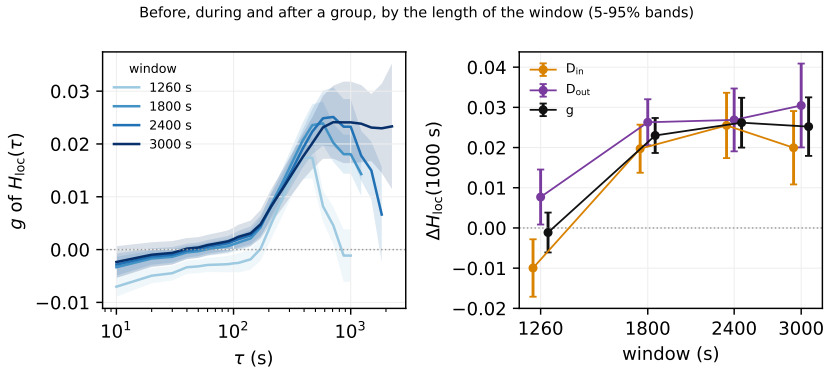


- g **excludes 0 for eight of nine quantities.** (Not for $H_{\mathrm{loc}}(1840\ \mathrm{s})$).
- The controls are near 0 for H_{loc}: time alone does not produce g.
- For climbing, time alone goes the other way.

Windows of 2400 s; 90% bands. Events: 1 595 (1 542 flights) for H_{loc}, 1 150 (1 113 flights) for the phase statistics.

→ also $d\,w$ and post − pre

g appears at every window length; $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is a fair quote from 1800 s



- $g > 0$ **for every window**. Plateau for Window $> 1800s$.

Events with the four windows: 3 549, 2 567, 1 595 and 695.

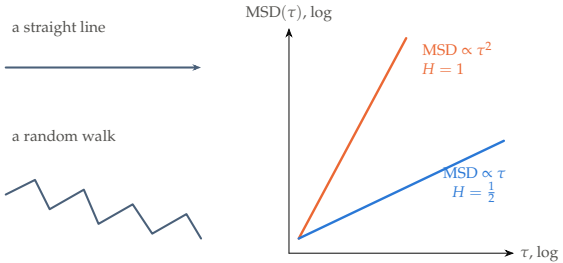
Take-home

1, 4	A group effect: in the raw data and within one flight	$+0.018$ $+0.026$ $H_{\mathrm{loc}}(1000\ \mathrm{s})$: raw, same flight
2	On the same route, group and solo fly alike	≈ 0 the group alone
3	The route carries the gap	96–98% of what the factors explain
+	More time in climb, in every design	$+3$ pp raw, same route, same flight

Reading: the group helps to find a better path; on the same path it adds little.

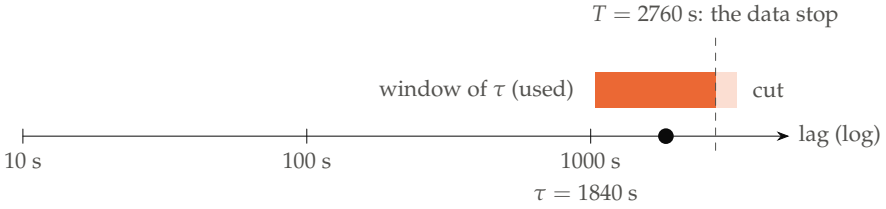
Appendix

How we compute H_{loc} in practice



- Per flight, the time-averaged MSD on the 10-s grid of its pieces; the ensemble MSD is the mean over flights, one weight per flight.
- $\text{MSD}(\tau) \sim \tau^{2H}$ on a geometric grid of lags; $H_{\text{loc}}(\tau) = \frac{1}{2} \times$ least-squares slope of $\log \text{MSD}$ against $\log \tau$ over the lags within ± 0.25 decades of τ (from $\tau/1.78$ to $1.78\,\tau$); with fewer than 3 lags, the 3 nearest.
- H: the same fit over all lags, from 10 s to the end of the dataset.

H_{loc} at the edge of the lags: a one-sided estimate



- At the longest lags T of a dataset the window is cut on its upper side: the slope uses only lags below T, so it is centred at a shorter lag than τ and is asymmetric.
- Example: $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on the dataset to $T = 2760$ s uses 1035 to 2760 s instead of 1035 to 3272 s; $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on the dataset to 1000 s uses 562 to 1000 s, but 562 to 1778 s on the dataset to 2760 s: complete.
- All curves of a dataset share T, hence the same cut: differences between populations are like for like, and every bootstrap draw is cut the same way, so the bands include it.

The phase statistics: how each flight counts

Example, glide speed = distance of glides / time in glides, for two flights:

flight	glide distance	glide time	flight time	glide speed
1	20 km	0.5 h	1 h	40 km/h
2	60 km	2 h	4 h	30 km/h

Ratio of sums. $\dfrac{20+60}{0.5+2} = 32$ km/h. Flight 2 weighs four times flight 1: it glides four times longer. The longest flights dominate.

Ours: average over flights, then ratio. $\dfrac{\text{mean}(20/1,\ 60/4)}{\text{mean}(0.5/1,\ 2/4)} = \dfrac{17.5}{0.5} =$ 35 km/h. A flight weighs by its share of time gliding, not by its length.

- With a unit u: $r_u = n_u/T_u$, $q_u = d_u/T_u$, stat $= \sum_u w_u r_u / \sum_u w_u q_u$; n_u, d_u summed over its runs, T_u its flight time.
- The MSD also gives one weight per flight. On the data: time in climb of S 32.9% (ours), 34.2% (ratio of sums).

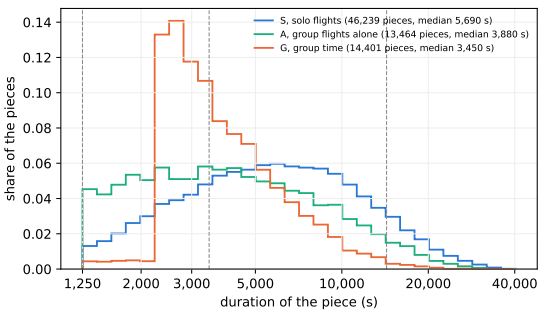
The seven phase statistics: formulas

Per flight, with T its flight time and sums over its runs of the phase; then averaged over flights as on the previous page.

statistic	formula	in words
time in climb (pp)	$100\,T_{\text{climb}}/T$	share of the flight spent climbing
time in search (pp)	$100\,T_{\text{search}}/T$	share spent looking for the next climb
searches → climb (pp)	$100\,N_{\text{found}}/N_{\text{search}}$	share of searches directly followed by a climb
climbs per hour	N_{climb}/T	climbs started per hour of flight
climb rate (m/s)	$\sum \Delta z_{\text{climb}} \,/\, T_{\text{climb}}$	height gained over time climbing
glide speed (km/h)	$3.6 \sum d_{\text{glide}} \,/\, T_{\text{glide}}$	distance over time gliding
height lost (m/km)	$1000 \sum h_{\text{lost}} \,/\, \sum d_{\text{glide}}$	height lost per km of glide

d_{glide}: straight distance from the start to the end of a glide (the chord); h_{lost}: height lost during it; N_{found}: searches whose next phase is a climb.

Datasets and the $\tau/0.8$ rule



A piece of ℓ s gives $(\ell - \tau)/10 + 1$ pairs of origins. Requiring $\ell \geq \tau/0.8$ gives the same share of origins at the longest lag.

- Dataset to T: pieces of at least $T/0.8$ s, at every lag up to T.
- 1000: pieces of 1250 s; 2760: 3450 s; longest (11 450 s): 14 312 s.
- Pieces of G are shorter: a group lasts at least $\tau^* = 38.1$ min.

Why the long dataset stops at 2760 s

caliper (month, hour, wing)	$T = 2250$	2760	3380
±1, ±1, class	311/395	166/231	74/111
±1, ±1, level	327/656	176/372	79/189
±2, ±2, class	513/725	**279/421**	129/228
±2, ±2, level	537/1095	294/650	139/370

pieces of G with a control, beginners/experienced; the rule asks for 200 in each level.

- $H_{\mathrm{loc}}(1840\ \mathrm{s})$ uses the lags within ±0.25 decades: 1035 to 3272 s.
- At $T = 2760$ the window is cut at 2760 s; at 3380 s no caliper reaches 200 pieces per level.

Why the difference $(G - S|G) - (A - S|A)$

Goal: separate the effect of the group from everything else group flights carry with them: the pilots and the day (its weather beyond season and hour).

$G - S|G$. Group time against solo controls on the same route, season, hour and wing. It holds the group, and what group flights carry with them: the pilots and the day.

$A - S|A$. The solo stretches of the same flights (same pilots, wings, days) against their controls. There is no group in A: what is left is the pilots and the day.

Difference. Subtract the second gap from the first:

$$\underbrace{G - S|G}_{\text{group} + \text{pilots, day}} - \underbrace{(A - S|A)}_{\text{pilots, day}} = \text{group}.$$

- At 1840 s: $+0.021$ for $G - S|G$, $+0.027$ for $A - S|A$, -0.006 for the difference: the gap is carried by the pilots and the day.
- It assumes pilots and day weigh the same in group time and alone, within the same flight.

How φ is computed

Time in climb (pp). One gap $\Delta(C) = \mathrm{G} -$ controls for each subset C of $\{R, T, W\}$ fixed; **fixing R moves it by an amount that depends on what is already fixed.**

fixed C	$\Delta(C)$
$\varnothing$	-9.55
R	$+2.20$
T	-10.35
W	-8.66
$R{+}T$	$+2.65$
$R{+}W$	$+2.64$
$T{+}W$	-9.29
$R{+}T{+}W$	$+2.96$

R added after	gap before $\to$ after	change	weight
nothing	$-9.55 \to +2.20$	$+11.75$	$1/3$
T	$-10.35 \to +2.65$	$+13.00$	$1/6$
W	$-8.66 \to +2.64$	$+11.30$	$1/6$
T, W	$-9.29 \to +2.96$	$+12.25$	$1/3$

Weight: how many of the six orders of fixing R, T, W put R in that state (2, 1, 1, 2 of 6).

- $\varphi_R = -\left(\frac{1}{3}\,11.75 + \frac{1}{6}\,13.00 + \frac{1}{6}\,11.30 + \frac{1}{3}\,12.25\right) = -12.05.$
- Same for T ($+0.19$) and W (-0.65), from the same eight gaps.

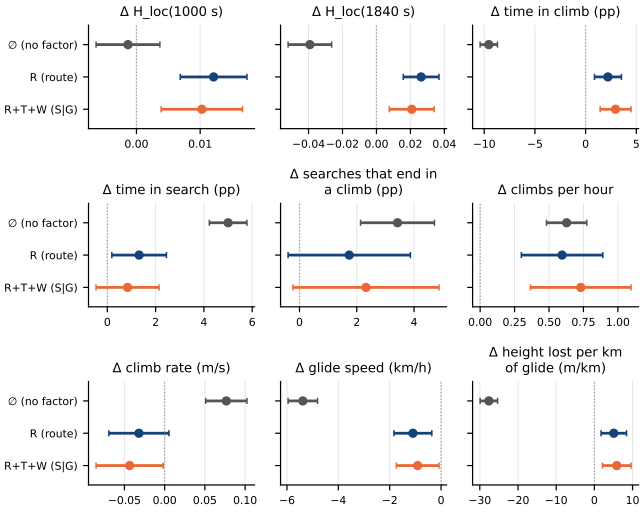
Why the three φ add up to the whole move

Fix the factors in one order: the three steps add up to $\Delta(F) - \Delta(\varnothing)$, whatever the order. Gain of each step, time in climb (pp):

order	gain of R	gain of T	gain of W	total
R, T, W	$+11.75$	$+0.44$	$+0.31$	$+12.51$
R, W, T	$+11.75$	$+0.32$	$+0.44$	$+12.51$
T, R, W	$+13.00$	-0.80	$+0.31$	$+12.51$
T, W, R	$+12.25$	-0.80	$+1.06$	$+12.51$
W, R, T	$+11.30$	$+0.32$	$+0.89$	$+12.51$
W, T, R	$+12.25$	-0.63	$+0.89$	$+12.51$
mean	$+12.05$	-0.19	$+0.65$	$+12.51$
$\varphi = -\text{mean}$	-12.05	$+0.19$	-0.65	-12.51

- Each order splits the same total among the three factors; the order changes only the split.
- φ_k: the mean over the six orders of the gain of k, opposite sign. The three means add up to the mean of six equal totals.
- So $\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W$ exactly: $-9.5 = +3.0 - 12.05 + 0.19 - 0.65$.

Same route, then all fixed: the nine quantities



90% bands. “No factor”: windows of long solo flights, not the raw S of slide 5.

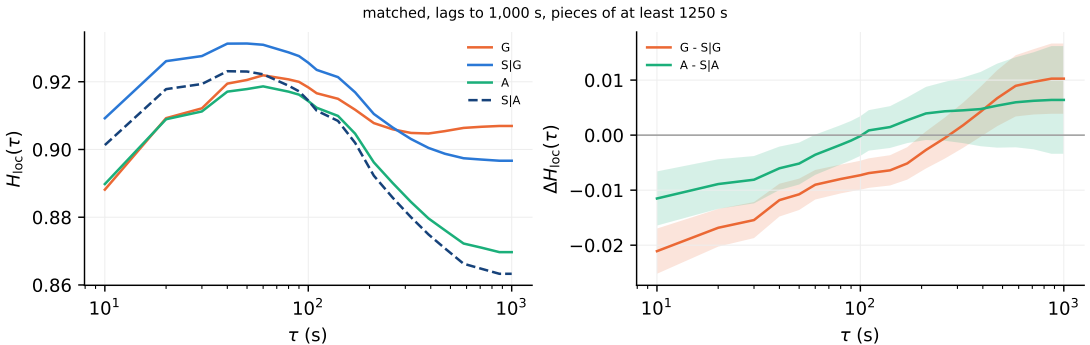
- G against three sets of solo controls:
  1. $\varnothing$, no factor fixed;
  2. R, same route;
  3. $R{+}T{+}W$, same route, season, hour and wing ($= \mathrm{S}\,|\,\mathrm{G}$).
- Fixing the route moves the gap, often across 0; **fixing season, hour and wing on top changes little.**

Matching: what is a control

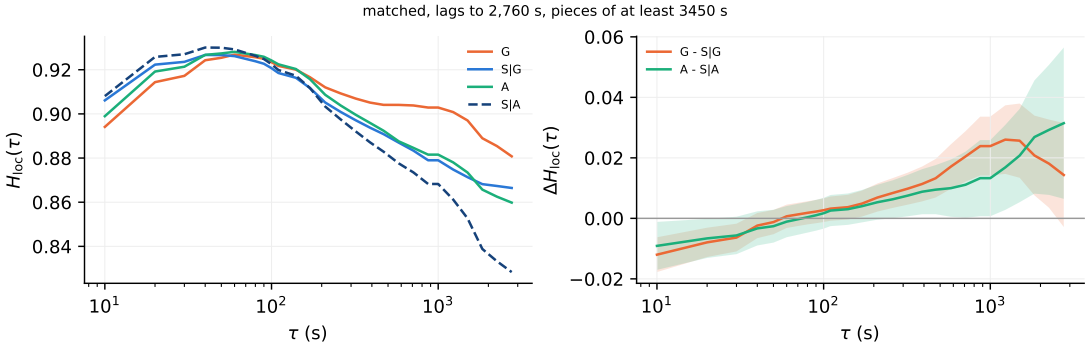
- Candidates: open flights with no contact of any length, on another date.
- Coverage: at least 80% of the points of the piece, one every 30 s, have a pool track point within 500 m, with the same direction.
- Calipers, tried in order: month ± 1 or ± 2, hour ± 1 or ± 2, wing class or level; the chosen one is the first with 200 pieces of G per level.
- Each unit is reweighted to the piece, so a piece with many controls counts once.

Isolation (nobody within 10 km) is off in the main result and is one of the robustness lines.

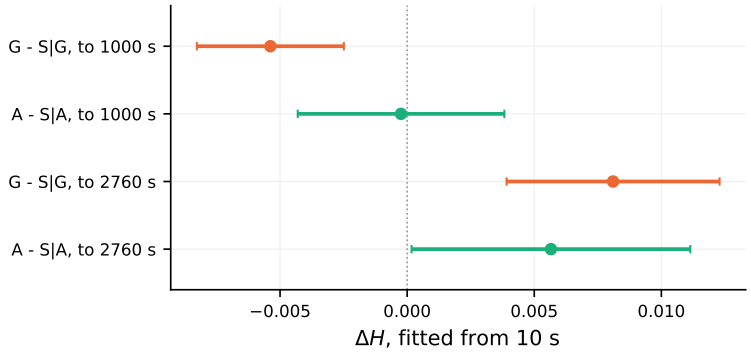
Same route, pieces of at least 1250 s: H_{loc} of G, S|G, A, S|A and their differences



Same route, pieces of at least 3450 s: the same; at 1000 s $\mathrm{G} - \mathrm{S}|\mathrm{G} = +0.024\ [+0.014, +0.034]$

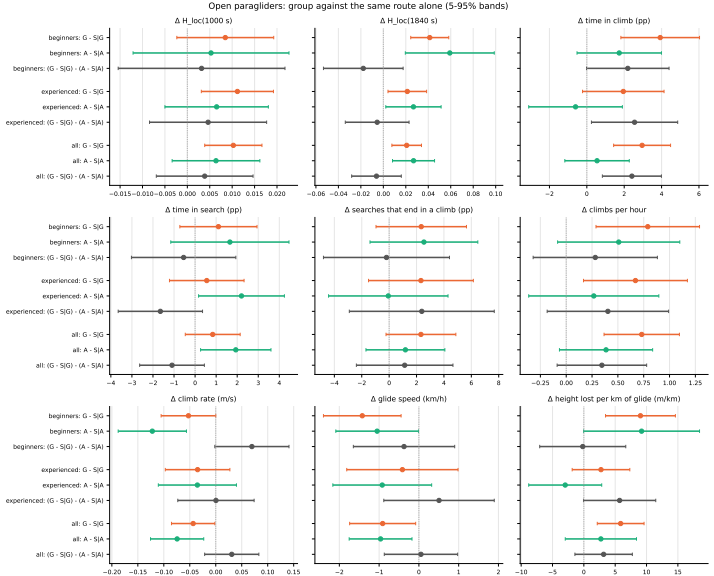


Fitted H on the same route

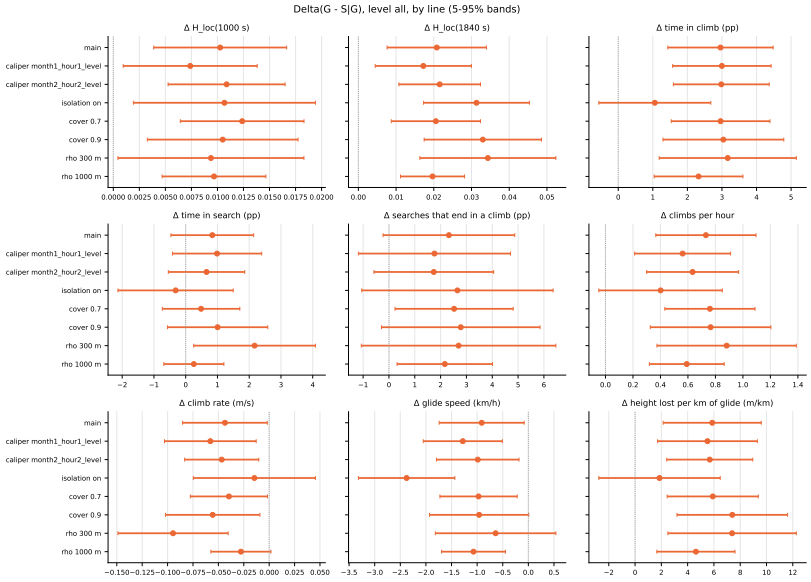


$G - S|G$: -0.005 $[-0.008, -0.002]$ to 1000 s and $+0.008$ $[+0.004, +0.012]$ to 2760 s. The sign depends on the lags: H mixes the small deficit at short lags and the gain at long lags.
$A - S|A$: about 0 to 1000 s, $+0.006$ to 2760 s.

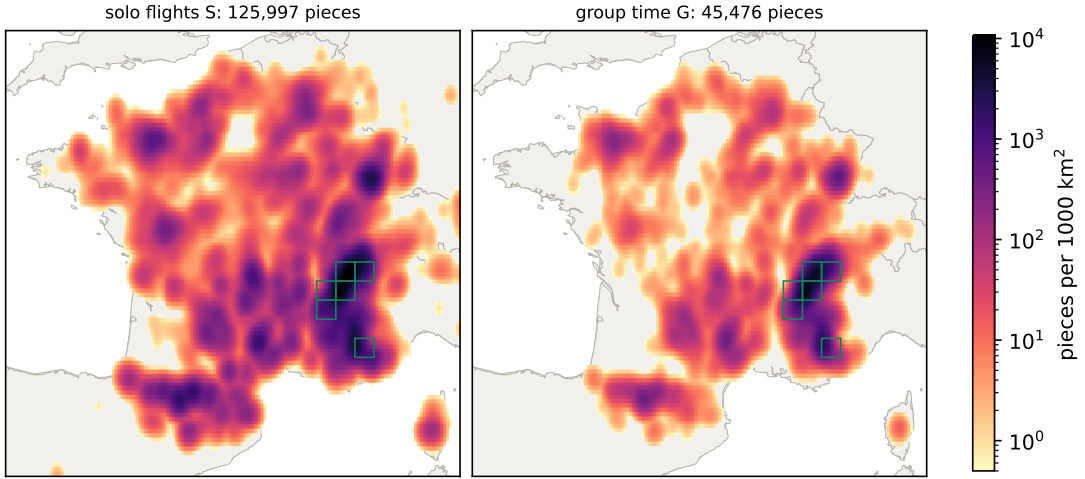
Same route: the nine quantities, by wing level



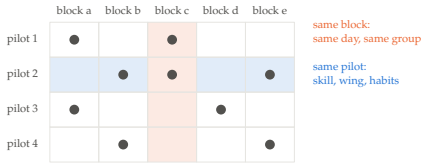
Same route, 8 lines: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ $+0.007$ **to** $+0.012$**;** $\Delta H_{\mathrm{loc}}(1840\ \mathrm{s})$ $+0.017$ **to** $+0.022$ **(**$+0.031$ **to** $+0.034$ **with a shorter long series)**



Where solo flights and group time are: pieces per 1000 km^2



Where the uncertainty comes from



- The archive is one sample: other seasons would bring other pilots and other days, and every number would change.
- Flights are not independent: those of one pilot share skill, wing and habits; those of one block share the day and the group.
- Pilots and blocks cross: a pilot flies in many blocks, a block holds many pilots.

Block. flights tied by an *episode* of the group identification (groups linked in time through a shared member), or by a *site-day* (takeoffs within 3 km on one day); chained into connected components.

A cell is one pilot in one block, about one flight.

The band: three bootstraps, one formula

- Draw the pilots with replacement; each unit weighs the count of its pilot; recompute the statistic. Its variance over the draws is V_P.
- Same for the blocks (V_B) and for the cells (V_C): 1 000 draws each.

Variance and band. units of one pilot or one block covary, so

$$V = \max\big(V_P + V_B - V_C,\ V_P,\ V_B\big), \qquad \text{band} = \hat{\theta} \pm 1.645\sqrt{V} \quad (90\%).$$

V_C is subtracted once: the pairs of one cell are in both V_P and V_B.

- All the numbers of a section use the same draws: a difference has the band of its own draws.
- Pieces and their controls are drawn together, control flights included: a control that serves many pieces moves them all at once.

Cameron, Gelbach and Miller, J. Bus. Econ. Stat. 29, 238 (2011).

Does the band hold? A check by simulation

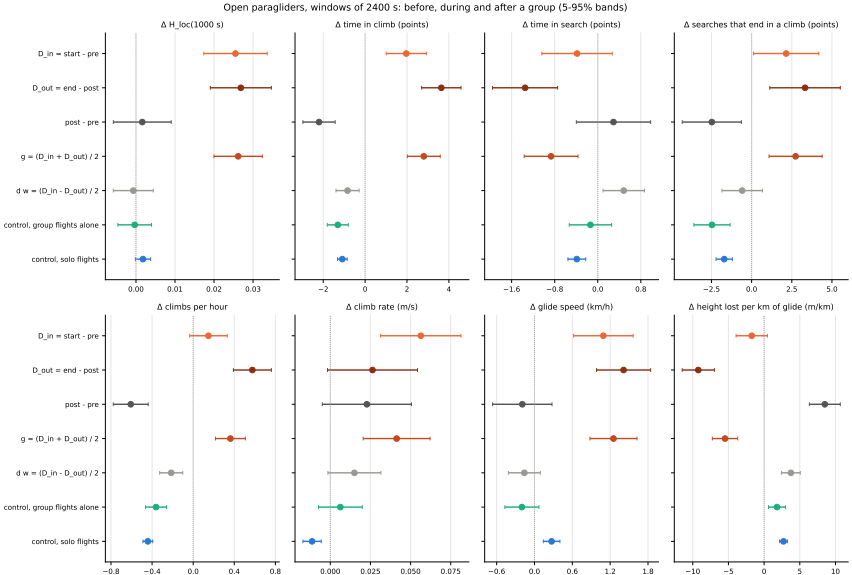
- On the real designs, replace every value by pilot + block + flight + unit effects: no group effect.
- A 90% band must hold 0 in 90% of the replicates.
- The two-way band does so in 88–99%; the bands used before did not (59% when pilots matter).

	raw		same route		same flight	
	new	old	new	old	new	old
unit only	91	90	99	100	90	88
flight only	92	91	89	100	90	90
half flight, half unit	88	85	92	100	88	88
pilot strong	88	59	88	100	89	81
block strong	88	84	88	99	89	88

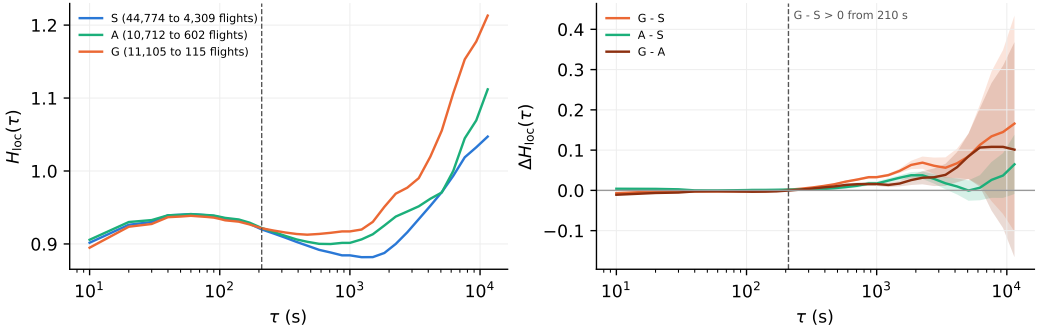
% of replicates whose 90% band holds 0.

Left out of the band: the rules of the matching (varied in the robustness lines) and the dependence on one year (planned).

Before and after, windows of 2400 s: all the differences

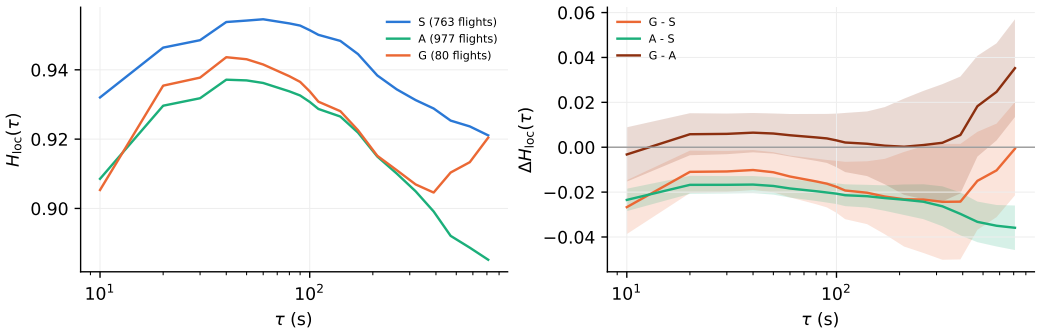


The same plots on a pool that shrinks with the lag



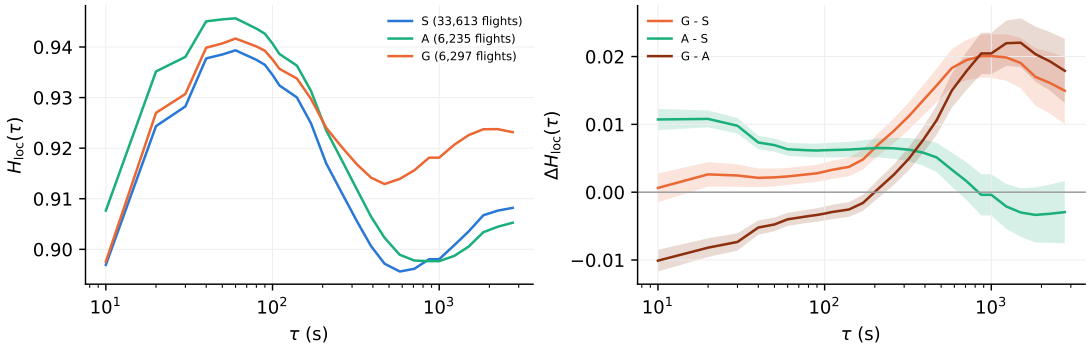
- At each lag the pieces of at least lag / 0.8: the flights fall from 44 774 / 10 712 / 11 105 at 10 s to 4 309 / 602 / 115 at 11 450 s (S / A / G).
- Beyond about 5000 s H_{loc} exceeds 1: the curve follows who is left, the longest flights, not the lag. This is why the main plots use the same pieces at every lag.
- G − S turns positive at 210 s, as in the main plots; the bands are very wide at the longest lags.

The same plots on short pieces only



- Pieces of 1000 to 1250 s, the same at every lag up to 800 s: 763 flights of S, 977 of A, 80 of G.
- Here G − S is negative, −0.01 to −0.02 up to about 300 s, and goes to 0 at 800 s: no gain of G on short pieces. With 80 flights the bands of G are wide.

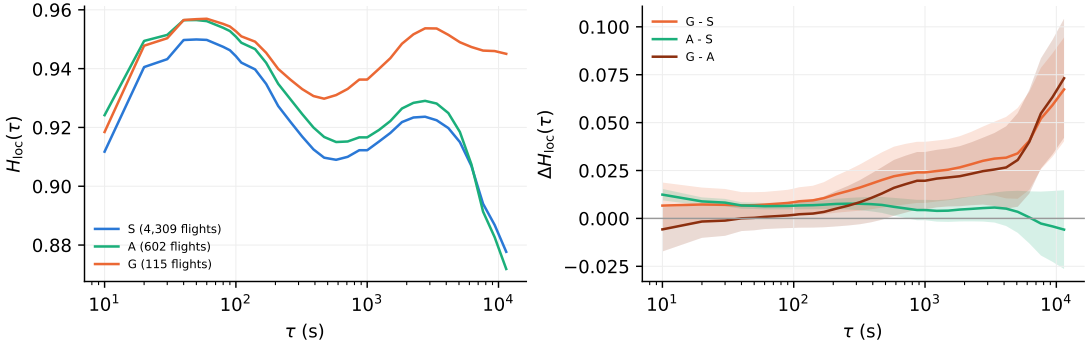
Raw at longer lags: dataset to 2760 s



Same flights at every lag: pieces of at least 3450 s, so lags up to 2760 s.

→ shrinking pool → short pieces

Raw at the longest lags: dataset to 11 450 s



- H_{loc} of S and A falls at the longest lags, that of G does not.
- G keeps few flights there: pieces of at least 14 312 s, the same at every lag up to 11 450 s.